

Global Insure Fraud Detection Case Study Report

1. Executive Summary

Insurance fraud is a critical concern for Global Insure, leading to substantial financial losses and impacting genuine policyholders. This case study explores how machine learning techniques—specifically Logistic Regression and Random Forest—can be used to detect fraudulent insurance claims using historical data. By leveraging robust data preprocessing, resampling, feature engineering, and model evaluation, this study demonstrates the ability to predict and flag potentially fraudulent claims with high accuracy. Both models were trained and validated to assess performance, understand key predictive features, and provide actionable insights.

2. Problem Statement

Insurance companies face challenges in detecting fraudulent claims that are often hidden within large volumes of legitimate transactions. Global Insure seeks to implement a data-driven solution to identify fraudulent patterns and predict future frauds effectively.

3. Data Overview

The dataset contains 1000 observations and 40 features, including customer demographics, policy information, claim details, vehicle specifications, and a target variable `fraud_reported` (Y/N). After data cleaning and preprocessing, both class balance and meaningful feature extraction were addressed.

- Redundant data, whether in the form of entire columns or specific values within columns, can negatively impact the performance, interpretability and efficiency of your machine learning models.
- Replaced null values with median/mode values and dropped columns with all null values. Replaced '?' with the category which has most count.
- Keeping them separate initially ensures correct transformations without unintended numerical operations on categorical data. Removed the columns with mostly unique values as it may be an identifiers like 'policy_number', 'incident_location', etc.

Fixing Datatypes: Machine learning models treat categorical and numerical features differently. Categorical variables often require encoding (one-hot or label encoding), whereas numerical features can be used directly in computations.

- Change the data type of 'incident_date' to 'datetime64[ns]'.
 - Handling illogical and negative values.
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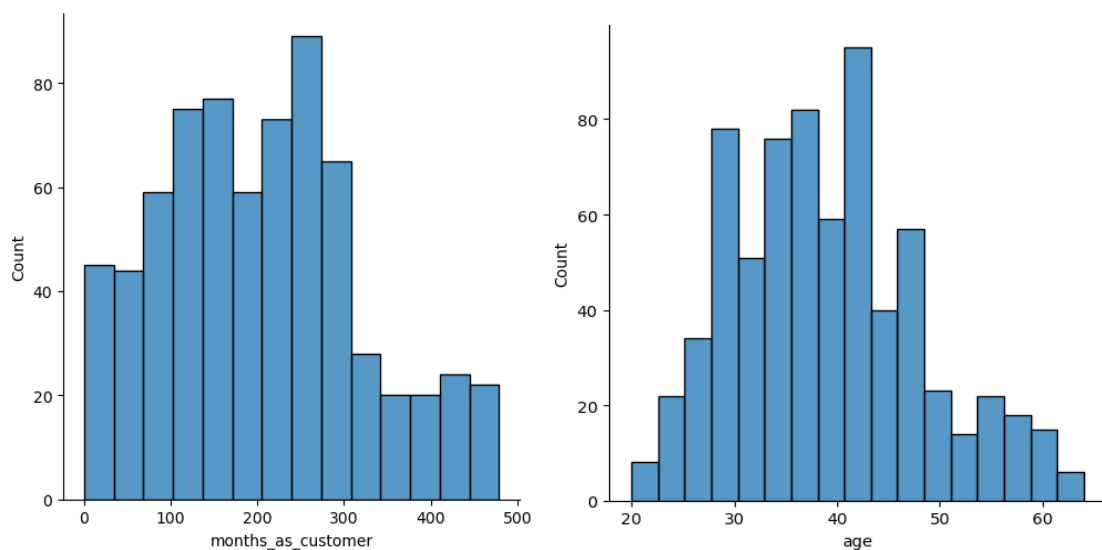
4. Train-Validation Split

- Separating the dependent and independent variables.
- Splitting training and the validation data with 70:30 ratio.
- Resetting the index of all the data frames.

5. EDA on Training data

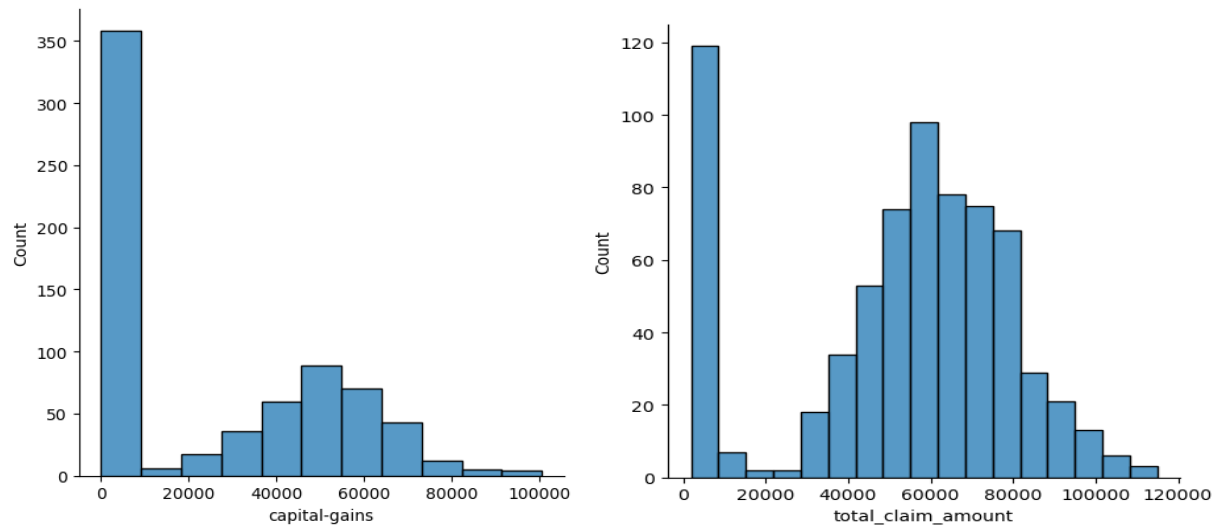
5.1 Univariate EDA Summary

- Months as Customer
 - Distribution is right-skewed, with most customers having been with the company for 100 to 300 months.
 - A few outliers exist with very long tenures, possibly loyal long-term customers.



- Age
 - Age is mostly concentrated between 30 and 45 years.
 - Fewer younger (<25) and older (>60) policyholders, suggesting a mid-age customer base.

- Capital Gains
 - Strong right-skew: majority have no capital gains.
 - Small proportion of customers have significant gains, indicating high variability.
- Total Claim Amount
 - Nearly normal distribution but with a peak at lower claim amounts (~\$0).
 - Majority of claims lie between **\$30,000 and \$80,000**, but some large claims exist.



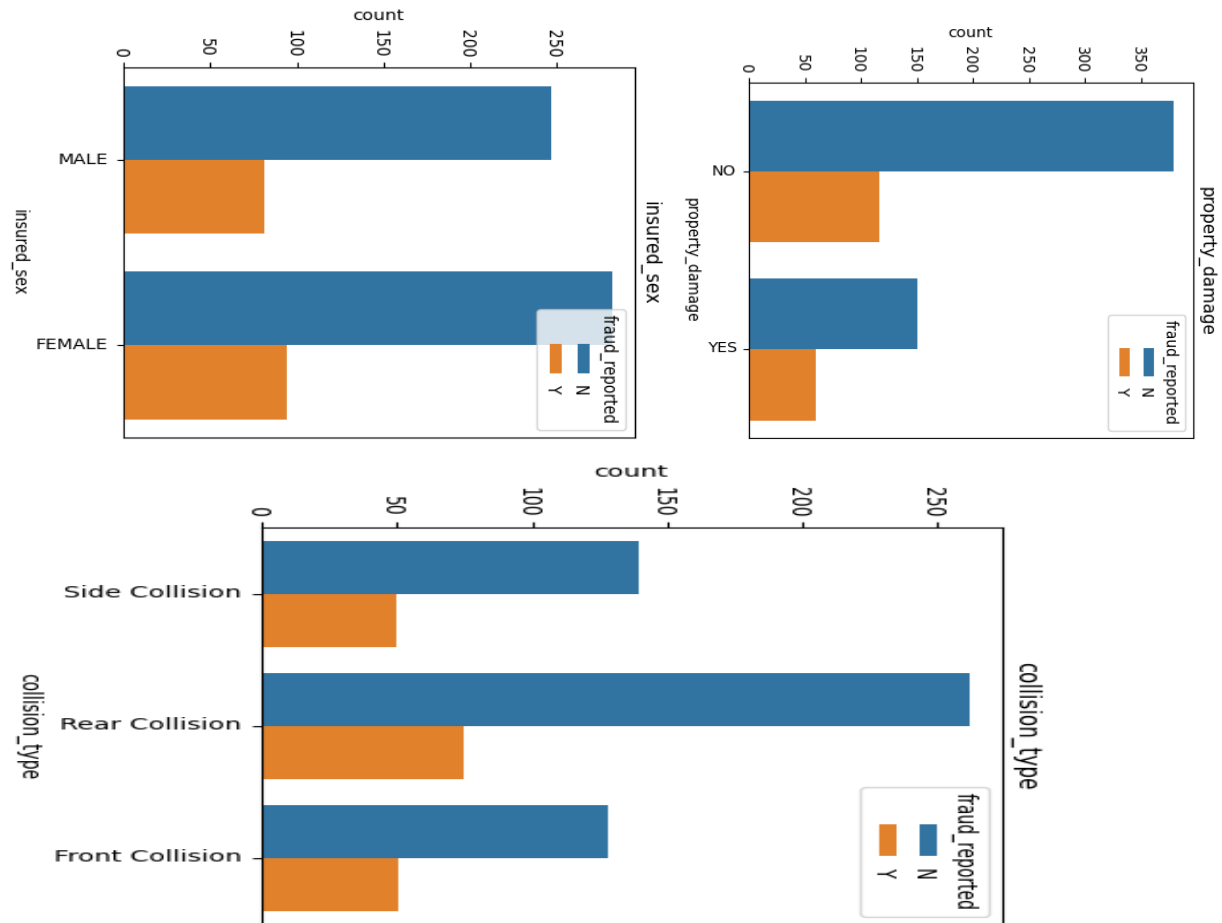
5.2 Bi-variate EDA Summary

1. insured_sex vs fraud_reported

- Observation: Both males and females have reported fraudulent claims.
- Insight: A slightly higher number of fraudulent claims were reported by females compared to males, despite similar proportions of fraud in both groups.
- Implication: Gender does not show a strong predictive pattern for fraud on its own, but it's still worth including in the model.

2. collision_type vs fraud_reported

- Observation: Rear collisions have the highest overall number of claims, including frauds.
- Insight: Fraud is relatively evenly spread across all collision types, though rear collisions slightly lead in fraud count.
- Implication: Collision type may help understand fraud tendencies when combined with other factors.



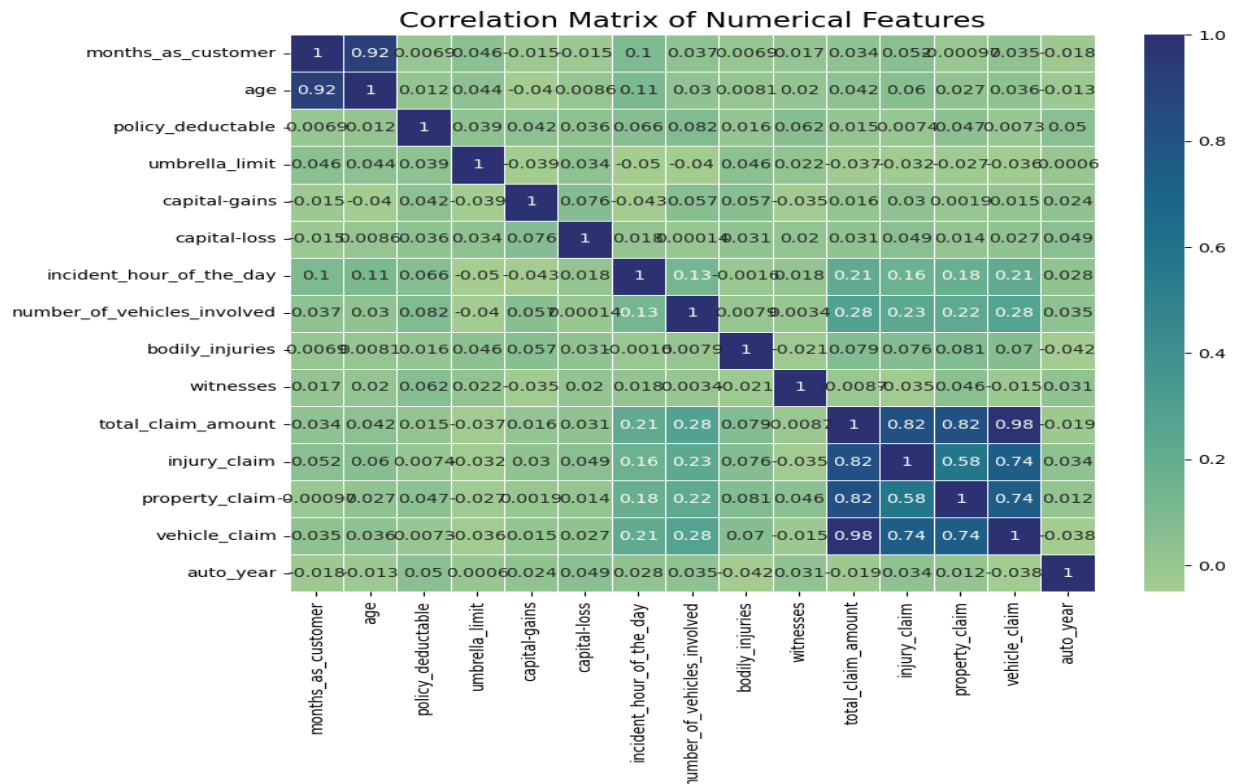
3. property_damage vs fraud_reported

- Observation: Most fraudulent claims were reported when property damage was marked as 'NO'.
- Insight: This could suggest attempts to hide or manipulate claim facts — i.e., claiming injuries or losses without physical property damage.
- Implication: This feature shows a strong signal and could be useful for flagging suspicious claims.

5.3 Multi-variate analysis:-

- months_as_customer and age have a strong positive correlation (0.92) — older customers have been with the company longer.
- total_claim_amount is highly correlated with vehicle_claim (0.98), injury_claim (0.82), and property_claim (0.82).

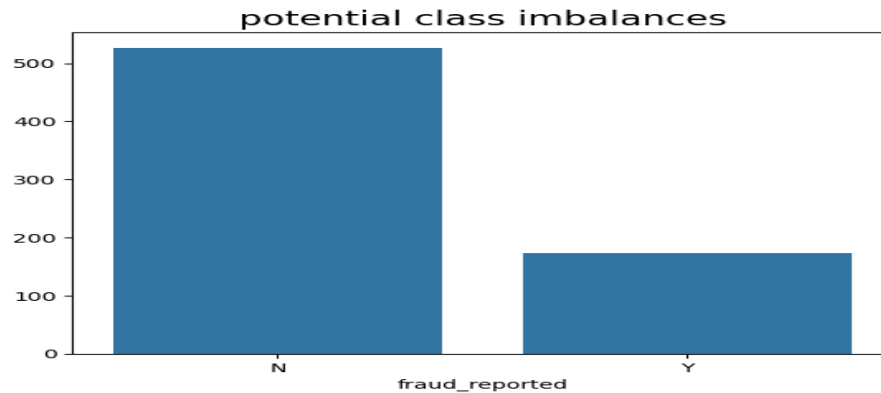
- injury_claim, vehicle_claim, and property_claim are also strongly correlated with each other — indicating multicollinearity risk.
- incident_hour_of_the_day and number_of_vehicles_involved show weak positive correlation with claim amounts.
- Variables like umbrella_limit, capital-gains, policy_deductable show little to no correlation with other features.



6. Feature Engineering

Resampling: Resampling addresses class imbalance, ensuring the model is not biased towards the majority class and can effectively learn patterns from all classes.

- RandomOverSampler was used to handle class imbalance. Using this method, you can add more observations from the minority class by replication. Although this method does not add any new information, there is no information loss.
- Original Distribution of target variable: $N=527$, $Y=173$
Resampled Distribution of target variable: $N=527$, $Y=527$



- This method increases the number of samples in the minority class by randomly duplicating them, creating synthetic data points with similar characteristics.

Feature Creation: Creating new features enhances the information available to the model, helping it capture complex relationships and patterns that may not be obvious from the original data. Well-designed features can introduce domain knowledge and improve the model's ability to distinguish between classes, leading to better performance..

- Interaction terms: `deductible_per_month`, `claim_per_vehicle`, `net_capital`, `claim_income_ratio`.
- Time-based features: `incident_month`, `incident_weekday`, `incident_week`
- Categorical simplification: Grouped occupations, education levels, and relationships.
- Additional derived features: `incident_time_category`, `total_injury_property_claim`, `total_vehicle_claim_ratio`.
- Removing the original columns, represented my new features.

Categorical Simplification & Dummy Creation:

- Merged low-frequency categories: Combine categories that occur infrequently or exhibit similar behavior to reduce sparsity and improve model generalisation.
- Applied one-hot encoding and binary mapping: Transform categorical variables into numerical representations using dummy variables for both training and validation data. Ensure consistent encoding between training and validation data.
- Scale numerical features to a common range to prevent features with larger values from dominating the model. Creating new features enhances the information available to the model, helping it capture complex relationships and patterns that may not be obvious from the original data. Well-designed features can introduce domain knowledge and improve the model's ability to distinguish between classes, leading to better performance.

Apply the same scaling to both training and validation data.

7. Logistic Regression Model

7.1 Feature Selection:

- Recursive Feature Elimination with Cross-Validation (RFECV) was used to retain top predictors based on ROC-AUC.
- RFECV is an extension of Recursive Feature Elimination (RFE) that automates feature selection by using cross-validation to determine the optimal number of features. It iteratively removes the least important features while refitting a model, ensuring that only the most relevant features are retained.
- Extends RFE by using cross-validation to automatically determine the best features, reducing overfitting risks but is computationally expensive

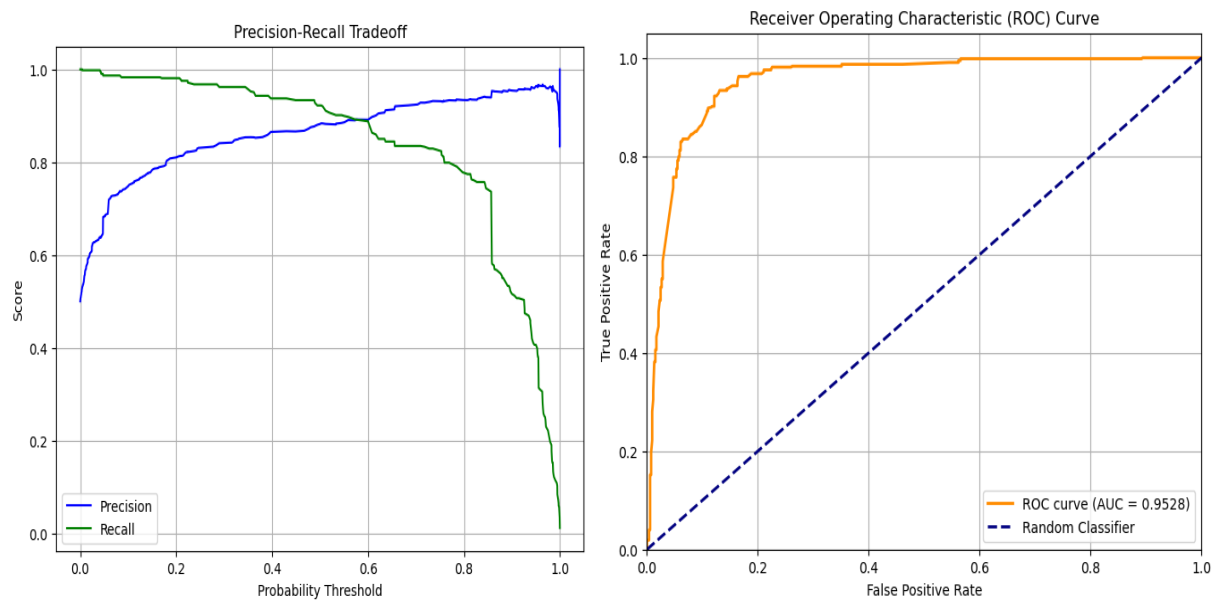
7.2 Model Building & Interpretation:

- Built using Statsmodels to evaluate statistical significance: The output summary table will provide the features used for building model along with coefficient of each of the feature and their p-value.
- The p-value in a logistic regression model is used to assess the statistical significance of each coefficient. Lesser the p-value, more significant the feature is in the model.
- Removed features with high p-values and multicollinearity (high VIF): VIF gives us the presence of multicollinearity and p-value gives significance of the feature in prediction.

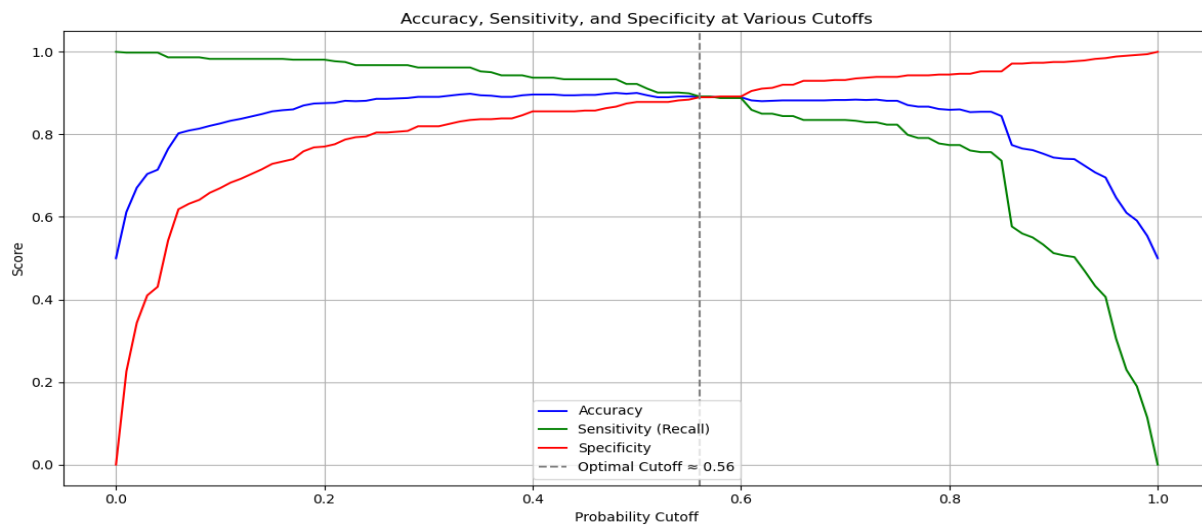
7.3 Evaluation (Training):

- Optimal Cutoff: 0.5 initially, then tuned for a base model.
- Accuracy: 0.9004, which is good as per standards.
- Final Evaluation after tuning:
 - Sensitivity (Recall) – 0.9222: The model correctly identifies 92.22% of actual fraud cases.
 - Specificity – 0.8786: It correctly identifies 87.86% of genuine (non-fraud) claims.
 - Precision – 0.8836: Of all claims flagged as fraud, 88.36% were actually fraudulent.
 - F1 Score – 0.9025: This balances precision and recall, showing strong overall performance.

In short: The model is highly effective at catching fraud while maintaining good accuracy in not flagging genuine claims.



- The following graph helps in finding optimal cutoff for the regression model using accuracy, sensitivity and specificity values.



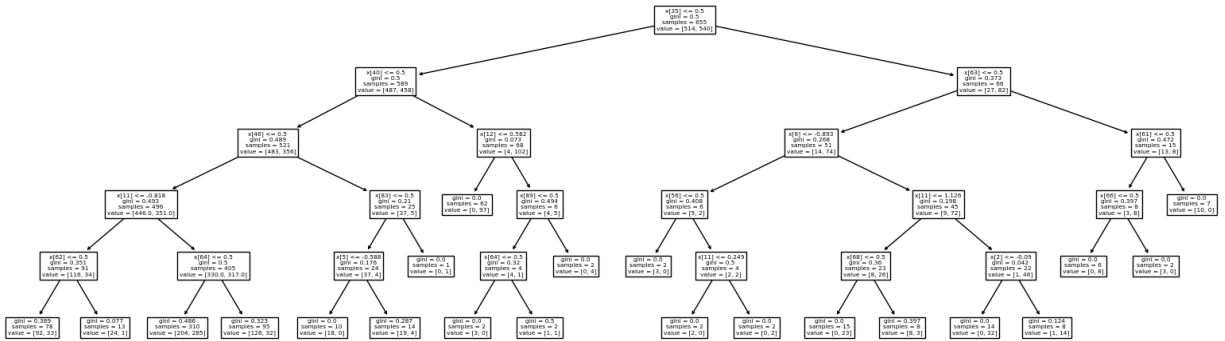
8. Random Forest Model

8.1 Feature Importance:

- Identified and retained most influential features using RandomForestClassifier

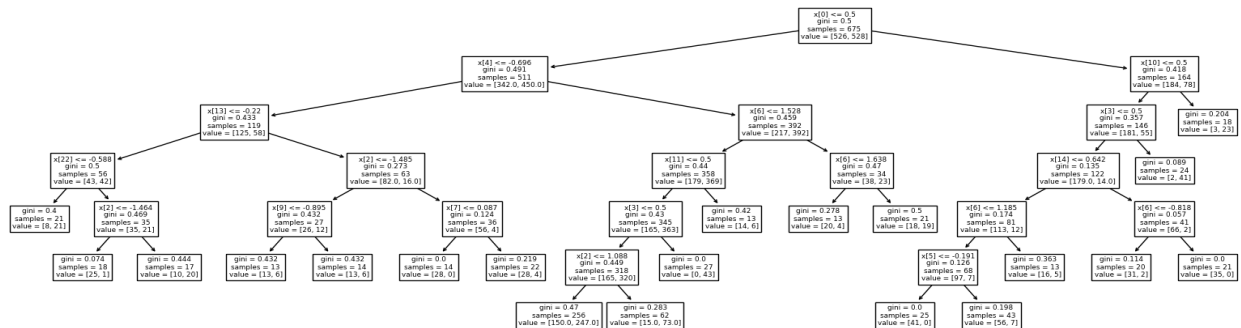
8.2 Base Model:

- Accuracy: 1.0000 (overfitting suspected)
- Cross-Validation Accuracy: Mean = 0.9326, Std Dev = 0.0118.
- Sample base Tree model :-



8.3 Hyperparameter Tuning: Hyperparameter tuning optimises a model's learning process, helping prevent overfitting and underfitting while improving performance on the given data. Since hyperparameters control the model's learning behaviour, fine-tuning them can significantly impact accuracy, precision, recall or other key metrics.

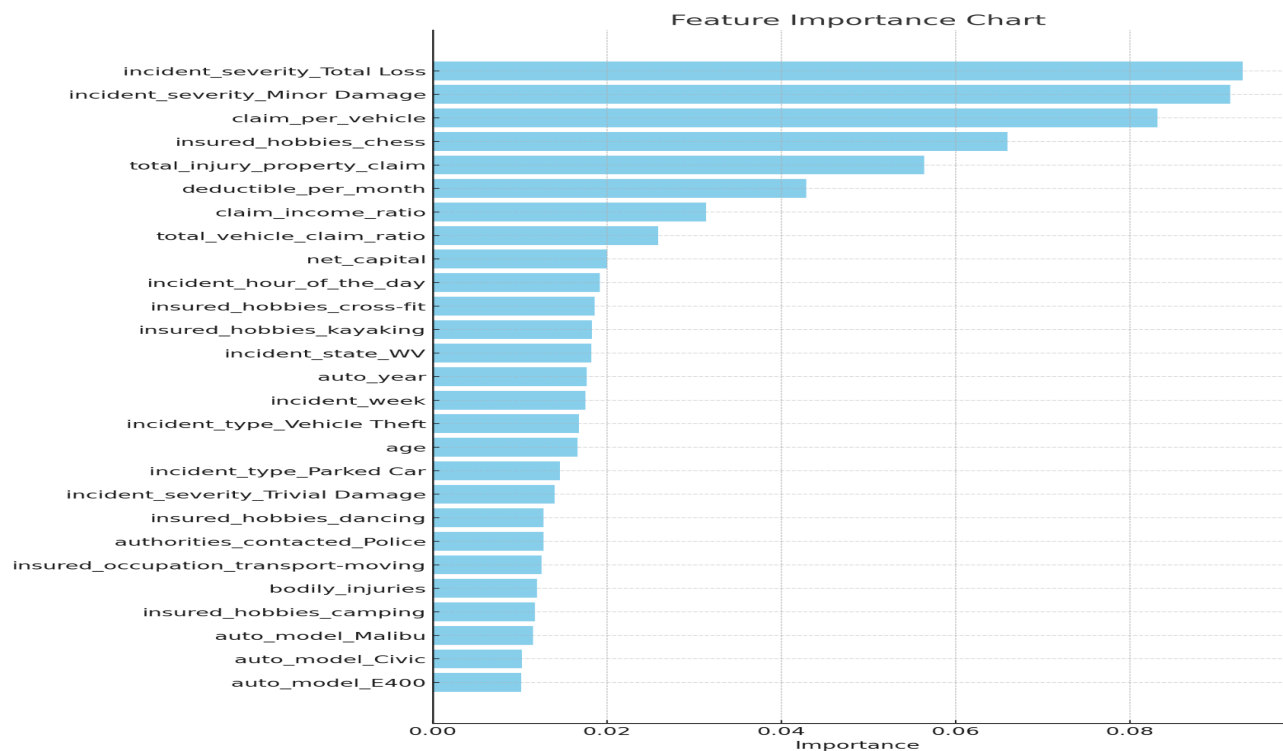
- Best Params: max_depth=6, max_features=6, min_samples_leaf=13, n_estimators=29
- Post-tuning Accuracy: 0.8776, The model correctly predicts about 88% of all claims.



- Final Metrics:
 - Accuracy - 0.8776: The model correctly predicts about 88% of all claims.

- Sensitivity – 0.8861: It identifies 88.61% of fraudulent claims correctly.
- Specificity – 0.8691: It correctly identifies 86.91% of genuine claims.
- Precision – 0.8713: Of the claims predicted as fraud, 87.13% are actually fraudulent.
- F1 Score – 0.8786: Indicates a good balance between precision and recall.

In short: After tuning, the model performs reliably across all key metrics, showing it's well-calibrated for fraud detection without overfitting.



9. Validation Evaluation

9.1 Logistic Regression on Validation Data:

- Accuracy (0.83): The model correctly predicted 83% of the cases.
- Sensitivity (0.7027): It correctly identified ~70% of the actual fraud cases.
- Specificity (0.8717): It correctly identified ~87% of the genuine claims.

- Precision (0.6420): About 64% of the claims predicted as fraud were truly fraud.
 - F1 Score (0.6710): Balanced performance, but slightly weaker in identifying fraud compared to Random Forest.
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9.2 Random Forest (Validation Data):

- Accuracy (0.83): Same overall prediction accuracy as Logistic Regression.
- Sensitivity (0.8108): Better at detecting fraud (81% of fraud cases identified).
- Specificity (0.8363): Slightly lower than logistic regression in detecting genuine claims.
- Precision (0.6186): Slightly lower precision, but better recall.
- F1 Score (0.7018): Overall better balance in fraud detection than Logistic Regression.

Summary:

- While both models have the same accuracy, Random Forest does a better job identifying fraudulent claims (higher recall and F1 score), making it more reliable for fraud detection tasks.
 - Also the difference between training and validation score is in acceptable range.
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10. Business Insights

Q1: How can we analyse historical claim data to detect patterns that indicate fraudulent claims?

By transforming and enriching historical claim data through feature engineering and simplifying categorical variables, we reveal underlying patterns (e.g., higher claim/income ratio, specific hours of incidents, certain occupations) that are strong indicators of fraud. ML models then learn from these patterns to identify future suspicious claims.

Q2: Which features are most predictive of fraudulent behaviour?

Key features included:

- `claim_per_vehicle`
- `claim_income_ratio`
- `incident_hour_of_the_day`

- insured_occupation
- policy_csl
- auto_make and auto_model

These features were repeatedly selected by RFECV and ranked highly in feature importance scores.

Q3: Can we predict the likelihood of fraud for an incoming claim, based on past data?

Yes. Both models achieved over 83% accuracy on validation data, confirming their ability to generalize predictions. With continuous learning from new claims, the prediction accuracy can be further improved.

Q4: What insights can be drawn from the model that can help in improving the fraud detection process?

- Claims reported late at night or from specific occupations had higher fraud probabilities.
- Higher ratios of claim to income or vehicle involvement hinted at inflated or fake claims.
- Simplifying features and regular audits of high-risk groups could preemptively reduce fraud attempts.
- Random Forests provided a transparent understanding of which features to monitor closely.

11. Conclusion

This study demonstrates that machine learning can significantly enhance fraud detection capabilities in insurance. By systematically engineering features and applying robust models, we successfully flagged potential frauds with high accuracy. Logistic Regression offers interpretability, while Random Forests bring strong predictive power. A hybrid approach, possibly incorporating both, could optimize results in production.

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